

# RapidBoxForest: Volumetric Reuse with Failure-Closed Route Status for Manipulator Motion Planning

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**Abstract**—Repeated manipulator planning queries in semi-static workcells require reusable kinematic evidence without reusing scene-local validity. RapidBoxForest (RBF) couples an axis-aligned joint-space box graph with a Lifelong Envelope Cache Tree (LECT). LECT gates obstacle-independent direct-store replay by semantic identity; the evaluated snapshot assumes a co-generated verified store. RBF derives link envelopes and refreshes scene labels and edges. Descriptor-aligned refinement and zero-gap exact-union coalescing construct a budgeted overlap graph with typed corridor transitions. A returned path is certificate-backed only when complete current-scene witnesses cover it; reported paths lack this provenance and are audited at a fixed step. A Shelf+IIWA compound comparison gives median builds of 0.171 s for warm affine-arithmetic interval forward kinematics (IFK-AA) replay and 2.770 s for on-demand five-level hierarchical interval forward kinematics (HIFK-5); the 1008.6-s prewarm is excluded. At method-specific points passing all catalog audits, RBF’s query-matched median path-length ratios span 1.00–1.05; bidirectional rapidly-exploring random tree (RRT-Connect) and Batch Informed Trees (BIT\*) span 1.02–1.36 and 1.04–1.07. Amortizing Build over ten queries lowers RBF’s mean build-plus-task diagnostic by 70–79% relative to one query. At 0.50 rad, HIFK-5 reduces median endpoint volume by 76% at 9.8 times the IFK-AA construction cost.

**Index Terms**—Motion planning, robot manipulators, multi-query planning, collision checking, configuration-space decomposition, link envelopes.

## I. INTRODUCTION

Manipulators repeatedly solve related queries in semi-static workcells, where kinematics stay fixed but start–goal pairs and local clutter change. A reusable planner must amortize free-space construction, separate persistent kinematic evidence from scene-local validity, and track whether every returned segment retains conservative current-scene support.

Existing methods occupy different reuse regimes. Probabilistic roadmaps, experience graphs, and safety certificates reuse connectivity, paths, or clearance neighborhoods, whereas tree and batch planners typically construct query-local search structures [1]–[4]. Certified polyhedral decompositions, safe-box planners, and Graphs of Convex Sets (GCS) methods instead make collision-free regions explicit [5]–[8]. This motivates the design question: how can explicit regions persist across queries and obstacle revisions while only identity-matched kinematic evidence is reused and every obstacle-dependent label is refreshed? Table I positions the proposed RapidBoxForest (RBF) and Lifelong Envelope Cache Tree (LECT) against representative planning families by reusable state and validation responsibility.

RBF addresses this design point by growing reusable axis-aligned configuration-space (C-space) boxes around scene-relevant seeds, validating them with envelopes of swept rounded links, and connecting accepted boxes in a typed corridor graph. Axis alignment trades general convex-region expressivity for dyadic addressing, inexpensive overlap/contact tests, and local invalidation. We evaluate this trade-off through build cost, fixed-step audit outcomes, path quality, and envelope tightness.

*a) Problem statement:* Let  $Q \subset \mathbb{R}^n$  be the joint-limit box of a fixed  $n$ -degree-of-freedom manipulator, and let  $\{\mathcal{O}^{(r)}\}$  denote workspace obstacle revisions under a fixed validation descriptor  $\delta$ . For revision  $r$ ,  $\mathcal{C}_{\text{free}}(\mathcal{O}^{(r)}) \subseteq Q$  contains the configurations free of active-link–obstacle collision under Assumption 1, and each query supplies start and goal configurations  $(q_s, q_g) \in Q^2$ . RBF constructs a finite family of policy-accepted C-space boxes and their adjacency graph for each revision. Across revisions, only obstacle-independent endpoint evidence whose joint-region key and recorded endpoint-evidence identity fields match may be replayed; the current  $\delta$  governs link-envelope derivation and scene validation, while box labels and graph connectivity are recomputed for  $\mathcal{O}^{(r)}$ . Within a revision, this query-agnostic scene structure is reused across start–goal queries, while query-endpoint attachment and repair remain query-specific. The target is a volumetric, auditable structure rather than an exhaustive decomposition of  $\mathcal{C}_{\text{free}}(\mathcal{O}^{(r)})$ .

*b) Reuse and certificate semantics:* Building on classical endpoint-to-link enclosures, RBF contributes a reusable planning architecture with versioned evidence identity, scene-local relabeling, typed transitions, and whole-route provenance. We distinguish *geometry evidence* (descriptor-keyed endpoint enclosures that may persist), *scene structure* (current boxes, labels, and graph edges), and *returned-path status* (certificate-backed only if current-scene containment and separation witnesses cover every segment; otherwise validation-required). We call accepted planning cells *policy-accepted regions* and reserve *certified free-volume regions* for conservatively validated boxes or finite-cover corridors. Table II states the full boundary.

The paper makes three contributions that jointly realize this contract:

- 1) *Two-level reuse with scene-local validation.* RBF reuses a typed box graph across queries within one scene. Across compatible revisions, LECT defines exact identity gates for obstacle-independent endpoint-evidence replay; RBF

TABLE I  
REUSABLE STATE IN REPRESENTATIVE PLANNING PARADIGMS

| Paradigm family                  | Reused state                                                              | Explicit reusable region cells  | Update or validation style                                              | Multi-query implication                                                               |
|----------------------------------|---------------------------------------------------------------------------|---------------------------------|-------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| Roadmaps and experience graphs   | Vertices, edges, path fragments, and cached checks                        | Method-dependent                | Recheck, retrieve, or repair affected graph elements                    | Strong connectivity or experience reuse; free-volume cells are not the standard unit. |
| Tree and anytime sampling        | Query tree or batch sample graph                                          | Not in the standard formulation | Primarily query-local; rebuilt or checkpointed                          | Fast discovery or quality improvement without persistent volumetric scene structure.  |
| Certified regions and safe boxes | Convex cells or safe boxes and their graph                                | Yes                             | Construct, certify, optimize through, or recertify regions              | Expressive or optimization-ready region reuse once a region collection exists.        |
| RBF                              | Within-scene boxes and typed graph; cross-revision LECT endpoint evidence | Yes                             | Relabel boxes; rebuild adjacency; replay only matched endpoint evidence | Two-level reuse with failure-closed route-status accounting.                          |

requires obstacle labels and graph connectivity to be recomputed under the current scene.

- 2) *Budgeted typed-region construction.* Descriptor-aligned refinement and zero-gap exact-union coalescing build a reusable certified-free (CertFree) overlap subgraph; a corridor layer augments it with typed tolerance-gap, segment, repair-witness, and finite-cover oriented bounding box (OBB) edges.
- 3) *Failure-closed route-status contract.* We formalize whole-route status from current-scene provenance: every endpoint attachment, bridge, repair, shortcut, or other segment must retain containment and separation support; otherwise the entire return is validation-required.

Algorithm 1 specifies reusable overlap-subgraph construction; Section V-C and Definition 1 define typed corridor augmentation and whole-route status; and Proposition 2 gives the conditional safety contract for identity-matched endpoint-evidence replay. Evaluation questions Q1–Q4 quantify replay, isolate envelope costs, and compare multi-query path-quality operating points.

Fig. 1 summarizes the method framework. Link-interval computation creates reusable geometry evidence. RBF growth recomputes scene labels and graph connectivity for the current obstacles, while LECT replay changes construction cost without changing validation responsibility. Sections III–VI make these blocks precise and then evaluate them.

## II. RELATED WORK

Region-based planners make collision-free volume an explicit planning object. IRIS (Iterative Regional Inflation by Semidefinite programming) constructs convex free-space regions, while configuration-space IRIS (C-IRIS) rigorously certifies C-space polytopes [5], [9]. Probabilistic nonlinear-programming and sampling-based extensions, including graphics processing unit (GPU) implementations, target faster region construction under probabilistic collision guarantees [10]–[12]. Visibility-graph clique covers seek compact approximate covers [13]. GCS optimizes paths through supplied region graphs, whereas GCS\* searches implicit GCSs [7], [14], [15]. RBF instead uses axis-aligned C-space boxes and a budgeted cover, prioritizing addressing, replay, and scene relabeling over general convex-region expressivity.

a) *Closest methods:* Marcucci *et al.* preprocess intersections in a supplied axis-aligned safe-box collection and then generate smooth paths within its union, whereas Morozov *et al.* precompute a cost-to-go lower bound for repeated online queries on a supplied GCS [6], [8]. Both begin from a supplied structure; RBF also constructs its budgeted box collection and specifies cross-revision persistence. Safety certificates cache clearance neighborhoods; dynamic roadmaps update affected graph elements; Ashur *et al.* index swept-volume approximations of roadmap vertices and edges to refresh validity after discrete obstacle motions; Thunder repairs graph experience; and soft subdivision conservatively classifies cells [2], [3], [16]–[18]. RBF instead combines versioned joint-box endpoint evidence and current-descriptor link-envelope derivation with scene-local relabeling and failure-closed whole-route status.

b) *Endpoint bounds for continuous collision detection:* Swept-volume and interval-kinematics methods supply the enclosure primitives used here. Convex-hull generator bounds and capsule-radius lifts provide swept-set enclosures [19], [20]. The Proximity Query Package (PQP), Gilbert–Johnson–Keerthi (GJK) algorithm, and Flexible Collision Library (FCL) provide collision-query primitives, while interval Denavit–Hartenberg propagation bounds articulated kinematics [21]–[30]. Redon *et al.* combine such components for articulated-body continuous collision detection over time steps [31], [32]; Taylor-model and safety methods use related swept-sphere bounds [33]–[35]. These works provide the enclosure primitives used by RBF. Building on them, RBF keys joint-box endpoint evidence by an exact versioned identity, replays only matching records across revisions, derives link envelopes under the current validation descriptor, and recomputes current-scene separation.

c) *Graph reuse:* Reusable graph planners amortize effort across related queries. A probabilistic roadmap (PRM) amortizes a collision-checked roadmap across multiple queries, whereas LazyPRM delays node and edge checks until they lie on a candidate path [1], [36]; adaptive cell methods split cells until they obtain a searchable accepted-cell graph [18], [37], [38]. RBF instead grows a budgeted overlapping cover; its vertices are policy-accepted regions, and transitions without containment provenance remain validation-required.

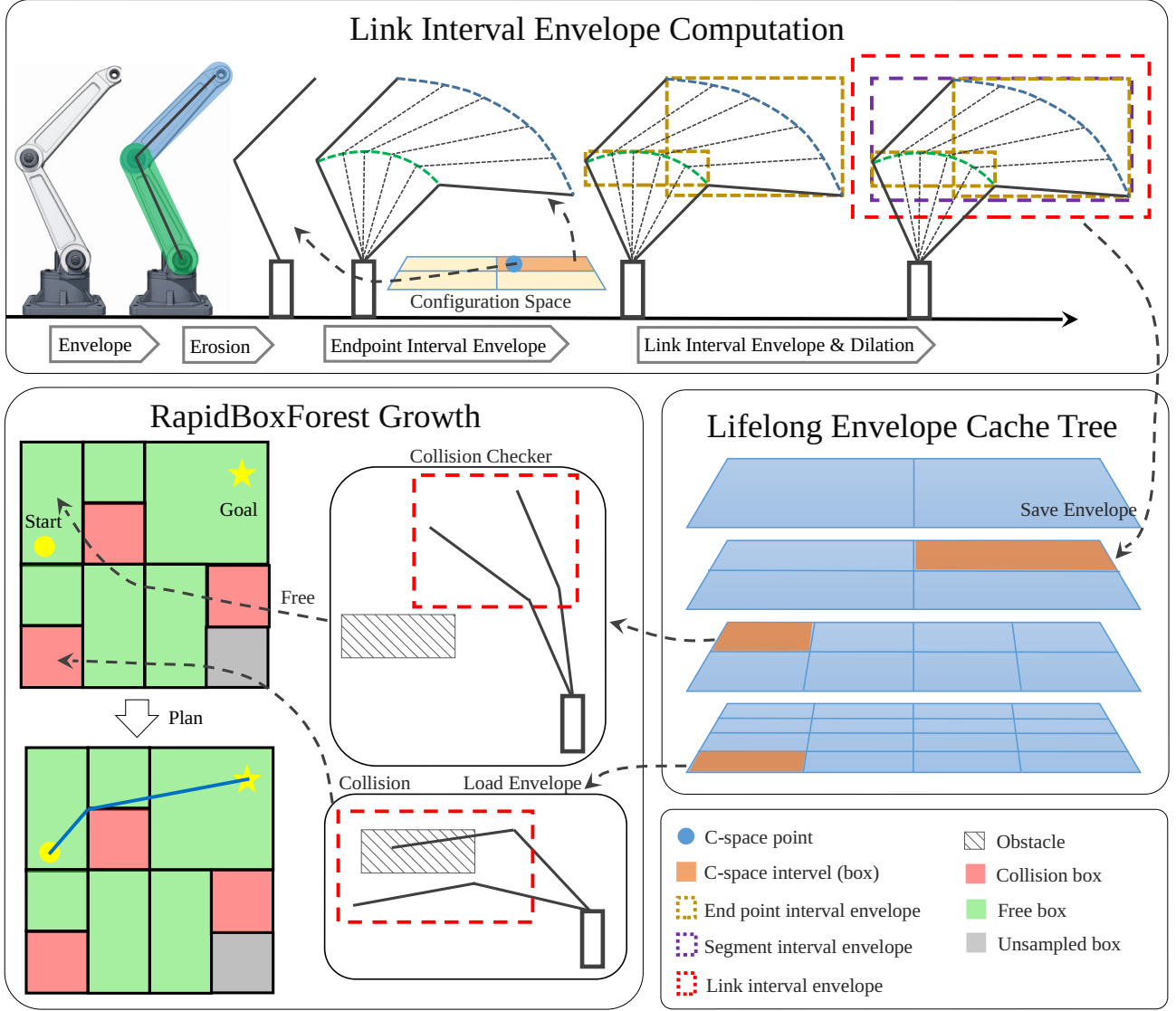


Fig. 1. RBF pipeline. Endpoint and zero-radius centerline envelopes induce radius-lifted link envelopes (top). The lower-left panel shows sample-guided growth with free, colliding, and unsampled boxes; statuses follow Table II. The lower-right panel shows LECT replay of endpoint evidence with recomputed obstacle labels and graph edges.

*d) Sampling and refinement:* Among rapidly-exploring random tree (RRT) variants, RRT-Connect is a bidirectional single-query planner. Batch Informed Trees (BIT\*) is anytime and asymptotically optimal over growing sample batches; Fast Marching Tree (FMT\*) is asymptotically optimal over a predetermined random sample set [39]–[41]. The Open Motion Planning Library (OMPL) provides benchmarking infrastructure [42], [43]; trajectory optimizers refine supplied paths [44]. RBF adds reusable volumetric context before repair, optimization, and final audit; it is not a single-query optimal planner.

*Synthesis.* Methods built on supplied regions and caches conditioned on obstacles retain different state. RBF uses a two-level split: a typed box graph is reused across queries within one scene, whereas only versioned, obstacle-independent endpoint geometry persists across compatible revisions. Current-scene labels and adjacency are rebuilt, and retained typed witnesses

separate search connectivity from certificate-backed connectivity.

### III. LINK INTERVAL ENVELOPES

This section defines endpoint and rounded-link envelopes for axis-aligned joint boxes and finite-cover OBB corridors; caching and graph construction follow in Sections IV and V. For a joint region  $I \subseteq \mathcal{Q}$ ,  $E_k(I)$  bounds endpoint  $k$ 's trajectory, and  $B_\ell(I)$  bounds the complete sweep of rounded link  $\ell$ :

$$I \mapsto \{E_k(I)\} \mapsto \{B_\ell(I)\}.$$

Axis-aligned boxes are the default primitive; local OBB corridors are certificate-eligible only after the finite-cover validation below. Geometry becomes certificate evidence only after conservative current-scene separation; empirical endpoint sources remain filters under Table II.

**Assumption 1** (Collision model for certificates). *The certificate model covers only workspace-obstacle collision between active rounded links and  $\mathcal{O}$ ; joint limits are enforced separately by  $I \subseteq \mathcal{Q}$ . The evaluated implementation represents  $\mathcal{O}$  as a finite union of closed workspace axis-aligned bounding boxes and treats boundary contact as collision. Each active-link capsule is the declared collision geometry or a closed outer enclosure of its link mesh. A certificate applies to one obstacle revision  $\nu_{\text{scene}}$ : the contract treats each edit as a new revision and requires all box and edge labels to be invalidated before the next query; concurrent scene mutation is excluded. Self-collision, attached bodies (including payloads), grippers, task regions, and other constraints lie outside this model and require an additional conservative validator and corresponding cache-identity fields. The experiments include only active rounded-link–workspace-obstacle collisions.*

**Definition 1** (Returned-path certificate status). *Fix a scene revision  $\nu_{\text{scene}}$ , its obstacle set  $\mathcal{O} = \mathcal{O}(\nu_{\text{scene}})$ , the active-link model, and a validation descriptor  $\delta$ . We distinguish its endpoint-evidence fields, which record the endpoint source and split policy, from its current-validation fields, which record the link representation, active-link set and radii, and numerical policy. A construction witness for a continuous returned path  $\gamma : [0, 1] \rightarrow \mathcal{Q}$  consists of a finite partition  $0 = t_0 < \dots < t_m = 1$  and regions  $R_1, \dots, R_m$  satisfying  $\gamma([t_{j-1}, t_j]) \subseteq R_j$  for every  $j$ . Each  $R_j$  is either (i) an axis-aligned box whose descriptor-matched link envelopes pass the current-scene conservative separation test, (ii) a zero-gap exact finite union of such boxes under the same revision and descriptor, or (iii) a C-space OBB whose complete region passes the conservative affine-zonotope endpoint/link validator under the same recorded numerical policy. The reporting contract assigns certificate-backed status to  $\gamma$  if and only if the implementation retains such a finite construction witness for the entire returned path. The current implementation serializes neither complete route nor exact-union provenance; all reported RBF returns are therefore validation-required.*

Certificate-backed status is construction-based and distinct from discrete audit; postprocessing must retain the complete witness in Definition 1.

#### A. Geometric Foundation

Let  $C_0$  be a body-frame shape. The rigid transform  $T(q)x = R(q)x + t(q)$  maps a body-frame point  $x$  to workspace at configuration  $q$ , with rotation  $R(q)$  and translation  $t(q)$ , and  $S_I(C_0)$  denotes the workspace set swept by  $C_0$  as  $q$  ranges over  $I$ . Let  $\mathbb{B}_p(r)$  denote the radius- $r$  ball in the  $p$ -norm, and let  $\oplus$  denote Minkowski sum. For a capsule  $C = S \oplus \mathbb{B}_2(r)$  with zero-radius center segment  $S$ , the fixed-radius ball commutes with rigid motion:

$$S_I(C) = S_I(S) \oplus \mathbb{B}_2(r).$$

Thus a rounded link can be bounded by first bounding the zero-radius center segment and then lifting by the link radius. For a convex body  $P = \text{conv}(v_1, \dots, v_m)$  with vertices  $v_i$ ,

outer bounds on the vertex trajectories also bound the swept body,

$$S_I(P) \subseteq \text{conv}\left(\bigcup_{i=1}^m \widehat{\Gamma}_i(I)\right).$$

Here  $\widehat{\Gamma}_i(I)$  conservatively bounds the workspace trajectory of vertex  $v_i$ ,  $i = 1, \dots, m$ , over  $I$ . A capsule link needs only its two endpoint trajectories. Let  $\Gamma_a(I)$  and  $\Gamma_b(I)$  denote the exact workspace trajectories of the proximal and distal endpoints over  $I$ . RBF uses these swept-sphere bounds over a joint-space box and stores the result as reusable kinematics-keyed bound records.

**Proposition 1** (Endpoint envelopes induce link interval envelopes). *Consider a rounded link with body-frame endpoints  $a, b$ ,  $L = [a, b] \oplus \mathbb{B}_2(r)$ , and any joint region  $R \subseteq \mathcal{Q}$ . If endpoint envelopes  $E_a(R)$  and  $E_b(R)$  satisfy  $\Gamma_a(R) \subseteq E_a(R)$  and  $\Gamma_b(R) \subseteq E_b(R)$ , then*

$$S_R(L) \subseteq \text{conv}(E_a(R) \cup E_b(R)) \oplus \mathbb{B}_2(r).$$

*Consequently, any conservative outer representation of the right-hand side is a valid link envelope for the whole joint region  $R$ .*

*Proof.* For each  $q \in R$ , the moved center segment is the convex hull of its endpoints. Hence the union of these segments lies in  $\text{conv}(\Gamma_a(R) \cup \Gamma_b(R)) \subseteq \text{conv}(E_a(R) \cup E_b(R))$ . Adding the fixed-radius ball then yields the capsule sweep and the stated containment.  $\square$

**Corollary 1** (Conservative configuration-space box validation). *Let  $I \subseteq \mathcal{Q}$  be an axis-aligned configuration-space box and let  $\{B_\ell(I)\}$  be conservative link interval envelopes for all active rounded links. If  $B_\ell(I) \cap \mathcal{O} = \emptyset$  for every link  $\ell$ , then every configuration  $q \in I$  is collision-free with respect to the collision model in Assumption 1.*

*Proof.* By conservatism, each  $B_\ell(I)$  contains the complete workspace sweep of link  $\ell$  over  $I$ . Its disjointness from  $\mathcal{O}$  therefore excludes the modeled workspace-obstacle collision for every active link and every  $q \in I$  under Assumption 1.  $\square$

a) *Numerical conservatism:* Software soundness requires outward containment, positive descriptor-bound margins, and failure-closed separation; an inconclusive evaluation cannot accept a free region. The axis-aligned bounding-box (AABB) broadphase uses  $\mathbb{B}_\infty(r_\ell) \supseteq \mathbb{B}_2(r_\ell)$ , while the support-mapping representation (SupportHull) with GJK requires outward margins and failure-closed separation. Because intermediate floating-point operations are not machine-checked, certificate status is conditional on this recorded numerical contract.

#### B. Endpoint Interval Envelopes

For endpoint  $k$ , let  $\Gamma_k(I)$  be its exact trajectory over joint interval  $I$ . Any conservative set containing  $\Gamma_k(I)$  is an endpoint interval envelope; the basic record is an endpoint axis-aligned bounding box (AABB) because it is compact and directly reusable by all link-envelope representations; the implementation calls this record an interval AABB (iAABB). We

TABLE II  
RETURNED-OBJECT STATUS UNDER THE COLLISION MODEL IN ASSUMPTION 1 AND THE EXPERIMENTAL PROTOCOL IN SECTION VI

| Object or state                              | Status              | Condition                                                                                                                                                                                                    |
|----------------------------------------------|---------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Returned path with retained finite box cover | Certificate-backed  | A retained finite partition covers the entire returned path, and every traversed region passes conservative endpoint-to-link validation or is a zero-gap exact union of same-revision certified-free inputs. |
| Returned path with retained finite OBB cover | Certificate-backed  | A retained finite partition covers the entire returned path, and every path window lies in a C-space OBB region whose complete extent passes the conservative affine-zonotope endpoint/link validator.       |
| Gap, segment, or repair witness              | Validation-required | Certificate-backed only after conversion to a conservative box or OBB corridor.                                                                                                                              |
| Filter-derived path                          | Validation-required | Empirical or non-conservative evidence may guide planning but cannot certify continuous collision freedom.                                                                                                   |
| LECT replay record                           | No path claim       | Direct-store replay requires joint-region and endpoint-evidence identity matches; scene labels and graph edges are recomputed.                                                                               |

evaluate two conservative interval variants. Affine-arithmetic interval forward kinematics (IFK-AA) performs an unsplit interval pass along the Denavit–Hartenberg chain, with shared noise symbols preserving part of the trigonometric correlation. Hierarchical interval forward kinematics (HIFK) bisects the joint interval and hulls child endpoint boxes, reducing wrapping at higher materialization cost; its split depth is part of the cache descriptor. Critical-sample probes selected boundary and trigonometric candidates, while Analytical performs structured root search for low-order sinusoidal extrema. The reported versions lack an interval-closure proof and therefore serve as filters. The upper panel of Fig. 1 summarizes the endpoint-to-link conversion.

### C. C-space OBB Endpoint Envelopes

Axis-aligned boxes remain the global forest primitive; an OBB is local edge evidence for a bridge or repair. Let  $q_c$  be its center, let diagonal  $S \succ 0$  scale the joints, let  $U \in \mathbb{R}^{n \times m}$  have orthonormal columns in scaled coordinates, and let  $h \in \mathbb{R}_{\geq 0}^m$  contain the half-widths. With  $A = S^{-1}U \text{diag}(h)$  and  $\Xi = \{\xi \in \mathbb{R}^m : -1 \leq \xi_i \leq 1\}$ , the joint-space region is

$$R_{\text{obb}} = \{q_c + A\xi \mid \xi \in \Xi\}.$$

Certificate eligibility requires  $R_{\text{obb}} \subseteq \mathcal{Q}$ , verified containment of the represented path window, and conservative validation of the complete region; the fitted orientation alone provides no safety status.

a) *Finite path cover*: For a piecewise-linear candidate path  $\gamma$ , recursively selected windows  $J_\tau \subseteq [0, 1]$  and OBBs  $R_\tau$  must establish  $\gamma(J_\tau) \subseteq R_\tau \subseteq \mathcal{Q}$ . A successful cover satisfies

$$[0, 1] = \bigcup_{\tau=1}^M J_\tau, \quad \gamma([0, 1]) \subseteq \bigcup_{\tau=1}^M R_\tau \subseteq \mathcal{C}_{\text{free}}(\mathcal{O}).$$

Every complete OBB region must pass the correlated endpoint/link test below. An uncovered window, exhausted budget, or inconclusive test rejects the cover; sampled fallbacks remain search-only.

**Corollary 2** (Finite-cover OBB-path soundness). *Fix a scene revision  $\nu_{\text{scene}}$  and descriptor  $\delta$ , and write  $\mathcal{O} = \mathcal{O}(\nu_{\text{scene}})$ . Let  $[0, 1] = \bigcup_{\tau=1}^M J_\tau$  be a finite window cover of a continuous path  $\gamma$ , with  $\gamma(J_\tau) \subseteq R_\tau \subseteq \mathcal{Q}$ . Suppose that, for every*

*$\tau$ , conservative affine-zonotope endpoint enclosures induce conservative envelopes for all active links over the complete region  $R_\tau$ , and that those link envelopes pass current-scene separation under the same revision and descriptor. Then  $\gamma([0, 1]) \subseteq \mathcal{C}_{\text{free}}(\mathcal{O})$  under Assumption 1.*

*Proof.* For each  $\tau$ , Proposition 1 and current-scene separation give  $R_\tau \subseteq \mathcal{C}_{\text{free}}(\mathcal{O})$ . For any  $t \in [0, 1]$ , the window cover supplies an index  $\tau$  with  $t \in J_\tau$ , and hence  $\gamma(t) \in R_\tau \subseteq \mathcal{C}_{\text{free}}(\mathcal{O})$ . The claim follows.  $\square$

b) *Correlated endpoint envelope*: For  $q = q_c + A\xi$ , outward affine propagation gives

$$E_k(R_{\text{obb}}) = c_k + \sum_i g_{k,i} \epsilon_i \oplus [-\rho_k, \rho_k].$$

Here  $\rho_k$  outwardly encloses all nonlinear remainder; a sampled remainder is filter-only. For workspace direction  $d \in \mathbb{R}^3$ , its support is

$$h_{E_k}(d) = d^\top c_k + \sum_i |d^\top g_{k,i}| + \sum_{\mu=1}^3 |d_\mu| \rho_{k,\mu}.$$

The rounded-link support bound is therefore

$$\bar{h}_\ell(d) = \max\{h_{E_a}(d), h_{E_b}(d)\} + r_\ell \|d\|_2.$$

Coordinate directions provide the AABB broadphase; retained supports provide the SupportHull/GJK test. OBB records remain scene-local typed-edge evidence and do not become persistent LECT records or global forest nodes.

### D. Link-Envelope Representations

Endpoint records can be exported to multiple link representations without changing the upstream kinematic source. For a serial-chain capsule link  $\ell$  whose proximal and distal endpoint records are  $E_{\ell-1}(I)$  and  $E_\ell(I)$ ,  $B_\ell(I) = B_\ell^\circ(I) \oplus \mathcal{R}(r_\ell)$ , with  $B_\ell^\circ(I)$  bounding the centerline sweep and  $\mathcal{R}(r_\ell)$  denoting the selected radius-lift set for link radius  $r_\ell$ .

a) *AABB envelope*: AABB envelopes wrap the two endpoint envelopes in one box and pad by the link radius:

$$B_\ell(I) = \text{bbox}(E_{\ell-1}(I) \cup E_\ell(I)) \oplus \mathbb{B}_\infty(r_\ell).$$

Here  $\text{bbox}(\cdot)$  denotes the workspace AABB hull. Segment subdivision may retain an explicit union of padded sub-envelopes for more selective filtering; collapsing that union back to one box would lose the tighter bound.

b) *Fixed-direction slab envelope*: A  $k$ -direction discrete oriented polytope ( $k$ -DOP) stores radius-padded endpoint support intervals along fixed directions. After the AABB broad-phase, these slabs can reject candidates before GJK; only conservative separation can contribute to a CertFree decision.

c) *SupportHull envelope*: SupportHull stores support mappings for both endpoint enclosures and the link radius. The default implementation uses endpoint AABBs; correlated OBB corridors use the zonotope-plus-box records above. A bounded-work GJK narrow phase uses outward obstacle inflation, and an unresolved iteration limit returns Unknown rather than a free label.

Fig. 2 illustrates the resulting build, query, and update cycle on a two-link schematic.

#### IV. LIFELONG ENVELOPE CACHE TREE

LECT persists versioned, obstacle-independent endpoint evidence across compatible builds. RBF derives link envelopes under the current descriptor and refreshes every scene-dependent label; graph connectivity and query outcomes never cross scene revisions.

##### A. Cache Key and Split Policy

LECT is a dynamically grown binary  $k$ -d tree over  $\mathcal{Q}$ ; each node stores a joint interval, split metadata, and geometry records.

a) *Cache identity*: Replay requires a fixed base frame and an exact joint-region and endpoint-evidence identity match. The endpoint key binds robot kinematics, joint limits, tool frame, endpoint map, root domain, canonicalization and symmetry modes, split policy, endpoint descriptor, payload layout, schema, and builder version. Link radii are excluded; active links, radii, link representation, and numerical policy remain current-validation fields of  $\delta$ . For symmetry sector  $s$ ,  $\phi_s$  maps a native interval to its canonical interval and  $\Psi_s$  maps the cached enclosure back. Replay is allowed only if  $\Psi_s$  preserves outward containment; otherwise evidence is recomputed in native coordinates. The evaluated persistent store contains endpoint records only. Derived-link envelopes may be memoized within the current oracle, but are not replayed across revisions and carry no scene label.

Direct database replay checks the full semantic identity. The evaluated memory-mapped snapshot is assumed to be the immutable, co-generated image of its verified companion store; its header alone does not establish full semantic identity.

At schedule depth  $h$ , let  $\mathcal{S}_h$  be a bounded deterministic sample of intervals and  $I_d^-, I_d^+$  the midpoint children in dimension  $d$ . The depth-synchronous policy selects

$$d_h^* = \text{lex argmin}_d \left( u_d, \sum_{I \in \mathcal{S}_h} \max_{\eta \in \{-, +\}} \mu(I_d^\eta), d \right),$$

where  $d$  ranges over dimensions splittable for at least one  $I \in \mathcal{S}_h$ ,  $u_d$  counts unsplittable samples, and  $\mu$  is the descriptor-specific child score; an unsplittable child contributes its parent

score. For IFK-AA,  $\mu$  sums active-endpoint iAABB volumes; SupportHull uses its corresponding volume score. The tuple favors broadly splittable dimensions, then the smaller sum of per-sample worst-child scores, with  $d$  breaking ties. Ordered expansion and sampling make the schedule deterministic for a fixed root and descriptor; a step is defined only while the candidate set is nonempty. For a runtime interval unsplittable along  $d_h^*$  under the split policy, resolution uses the widest strictly positive-width allowed dimension, breaking equal-width ties by the smaller dimension index; if none exists, the interval is terminal.

b) *Sparse interval staging*: Planning may stage an exact interval without materializing every ancestor. If its endpoint record is absent, that interval is materialized before validation under the current descriptor.

##### B. Runtime Use and Validation Semantics

For a joint interval, LECT replays or materializes endpoint evidence, derives link envelopes under  $\delta$ , and applies the current-scene validator. A complete child partition may form a parent hull, but the parent still passes the same validator. Missing or inconclusive evidence yields Unknown. Cache hits may change enclosure tightness and hence the acceptance outcome, but not the meaning or validation obligations of any returned status.

**Proposition 2** (Current-scene safety from conservative replayed endpoint evidence). *Let  $I \subseteq \mathcal{Q}$  be an axis-aligned LECT joint box, and fix the base-to-workspace frame, current scene revision  $\nu_{\text{scene}}$ , and descriptor  $\delta$ . Assume that cached endpoint records conservatively enclose every active-link endpoint trajectory over  $I$ , that decoding—including any canonical-to-native mapping—preserves outward containment, and that the joint-region key and all recorded endpoint-evidence identity fields match. If link envelopes derived under the current-validation fields of  $\delta$  are conservative and all pass separation from  $\mathcal{O}(\nu_{\text{scene}})$ , then  $I \subseteq \mathcal{C}_{\text{free}}(\mathcal{O}(\nu_{\text{scene}}))$  under Assumption 1. This conclusion uses no label from an earlier scene. Replayed and freshly materialized records may differ and hence need not yield identical acceptance decisions.*

Identity matching establishes replay eligibility, not containment; conservative cached evidence remains an independent premise.

*Proof.* The replay premises preserve all endpoint inclusions. Proposition 1 therefore places each rounded-link sweep inside its derived conservative envelope. Current-scene separation and Corollary 1 give the stated inclusion.  $\square$

##### C. Storage and Revision Semantics

Persistent units are endpoint records; link envelopes are derived under the current descriptor and revalidated against the scene. Prewarming fills leaf records and derives ancestor hulls from complete child evidence, but imports no box or edge label. A scene revision invalidates all prior labels. Insertions and deletions may select boxes and edges for rechecking, but every retained or promoted object is relabeled only after

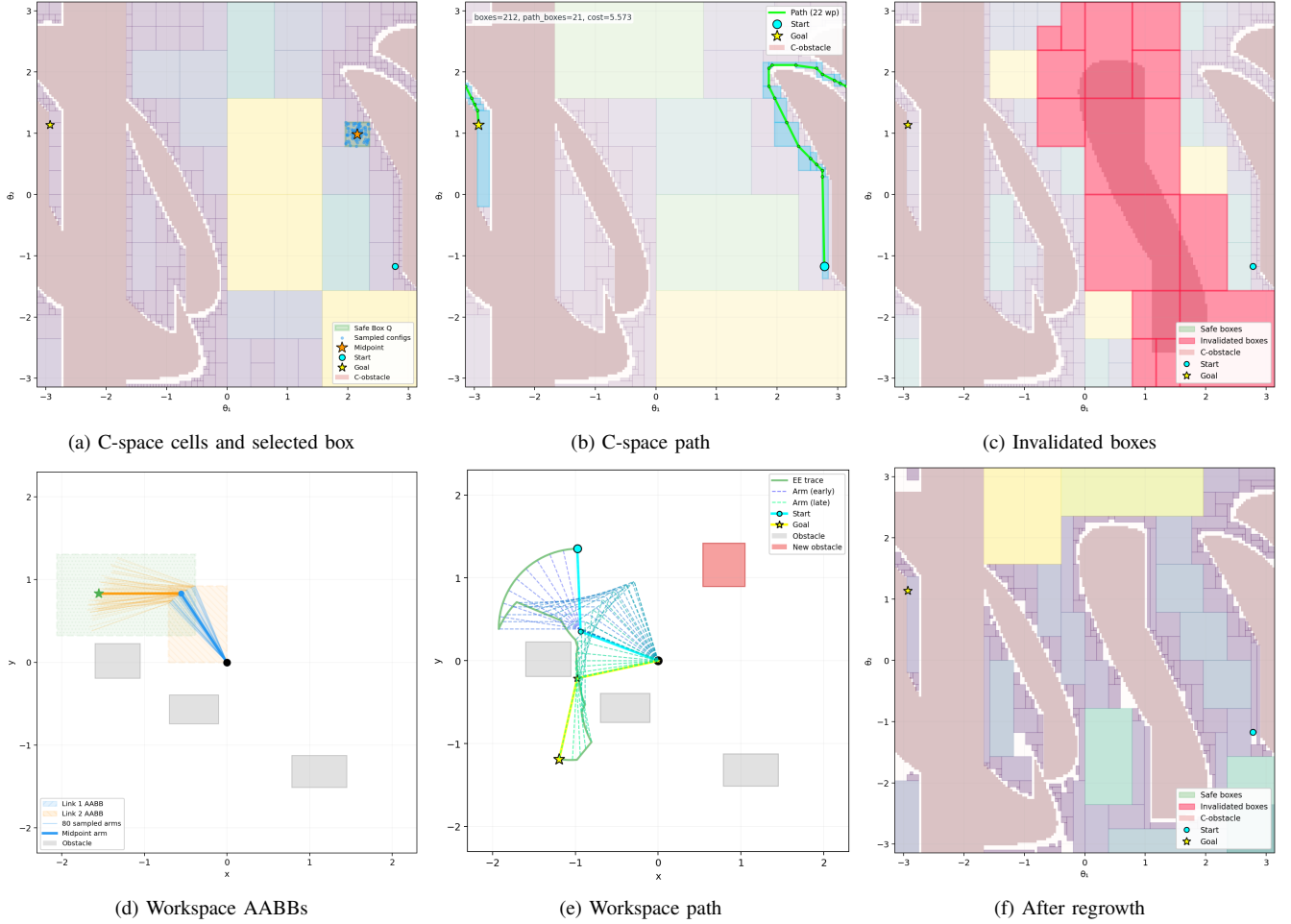


Fig. 2. Two-link RBF build-query-update schematic. Top: (a) selected C-space box, (b) route, and (c) invalidation; bottom: (d) workspace link AABs, (e) mapped route, and (f) regrowth. Embedded counts and costs are illustrative;  $\theta_0, \theta_1$  are joint coordinates and  $x, y$  are workspace coordinates.

current-scene validation; compatible LECT geometry remains reusable.

## V. RAPIDBOXFOREST CONSTRUCTION

Fig. 2 summarizes the planning layer. RBF turns envelope validation and LECT replay into a query-agnostic scene structure of scene-specific, policy-accepted C-space boxes and typed corridor adjacency. For a fixed obstacle set, later queries use box membership and graph search; LECT remains a separate evidence store. Under the construction contract, every accepted box is either checked against the current obstacles or inherits status through a same-revision, descriptor-matched exact-union witness.

A box is an axis-aligned joint interval  $B = [l^{(1)}, u^{(1)}] \times \dots \times [l^{(n)}, u^{(n)}] \subseteq \mathcal{Q}$ . A *policy-accepted box* has passed the configured scene-validation policy. Under the stated geometric and numerical assumptions, conservative link-envelope disjointness makes it eligible for CertFree; filter-only acceptance yields a validation-required planning region. Rejected cells are excluded from the forest, while selected rejected or unresolved domains are retained for local refinement or later update.

### A. Budgeted Construction Contract

Construction returns a budget-limited reusable cover rather than solving a maximum-volume, minimum-cardinality, or exhaustive-decomposition problem. Given the joint-limit box  $\mathcal{Q}$  and obstacle set  $\mathcal{O}$ , the scene policy  $V_{\mathcal{O}}(B; \delta)$  evaluates active-link envelopes for  $B$  under descriptor  $\delta$  and has range

$$V_{\mathcal{O}}(B; \delta) \in \{\text{CertFree}, \text{ProvisionalFree}, \text{Occupied}, \text{Unknown}\}.$$

The general accepted set admits the first two states; the evaluated strict-certificate build commits only CertFree cells. This status requires every active-link set  $B_{\ell}(B; \delta)$  to be a conservative outer envelope disjoint from  $\mathcal{O}$  under the recorded failure-closed numerical policy. ProvisionalFree denotes filter-only acceptance and remains validation-required; Occupied is a policy rejection, not a claim that all of  $B$  is in the configuration-space obstacle region; and Unknown denotes inconclusive or resource-limited evaluation. Rejected and unknown domains may be retained for refinement or update but never gain certificate status from a probe or discrete audit. The accepted region set of  $N$  boxes is

$$\mathcal{F} = \{(B_i, \sigma_i) : i = 1, \dots, N\}, \\ B_i \subseteq \mathcal{Q}, \quad \sigma_i \in \{\text{CertFree}, \text{ProvisionalFree}\}.$$

For a directly validated box,  $\sigma_i = V_{\mathcal{O}}(B_i; \delta)$ . For a finite group  $\mathcal{G}$  of CertFree boxes validated under one scene revision and descriptor, let  $H = \text{bbox}(\bigcup_{B \in \mathcal{G}} B)$ . The hull inherits CertFree only when the zero-gap set equality

$$\text{ExactUnion}(H, \mathcal{G}) \iff H = \bigcup_{B \in \mathcal{G}} B$$

holds. Connection tolerance  $\varepsilon_c$  is not used in this predicate; a tolerance-gap or nonexact hull requires direct whole-hull validation and is otherwise validation-required. In the reported runs, certificate inheritance is restricted to partition-native contiguous grid ranges, whose integer cell ranges establish the set equality; Q3 reports no coalescing. Tolerance-based merger paths are outside this certificate contract. The reusable output is  $(\mathcal{F}, G)$ , where  $G$  is the typed corridor graph of adjacencies and auxiliary transitions. The box budget targets fixed, query-independent anchors, frontiers, and connectors. “Forest” denotes rooted box-growth components before consolidation; the returned graph may contain cycles, and merged records need not retain ancestry.

For two boxes

$$B_i = \prod_{d=1}^n [l_i^{(d)}, u_i^{(d)}], \quad B_j = \prod_{d=1}^n [l_j^{(d)}, u_j^{(d)}],$$

let  $\varepsilon_c \geq 0$  denote the connection tolerance. RBF treats the boxes as candidate-adjacent when the closed intervals in every coordinate overlap or have a gap no larger than  $\varepsilon_c$ :

$$\max\{l_i^{(d)}, l_j^{(d)}\} \leq \min\{u_i^{(d)}, u_j^{(d)}\} + \varepsilon_c, \quad d = 1, \dots, n.$$

Only exact overlap or contact directly yields a contained box-corridor. Tolerance-gap and explicit segment edges are graph-search aids, not certificates, unless they are converted into conservative box-corridor or finite-cover OBB corridor regions; otherwise they require fixed-step validation.

### B. Build Pipeline

Algorithm 1 summarizes construction and certificate inheritance. It validates a descriptor-aligned frontier in batches, commits only CertFree cells, and coalesces only zero-gap exact unions of certificate-only inputs. Remaining maximal frontier cells enter an anchor-prioritized queue keyed by exact region identity.

Let  $\hat{V}_{\mathcal{O}}(I; \delta)$  obtain endpoint evidence, derive current link envelopes, and apply  $V_{\mathcal{O}}$ . Any unrecovered replay, materialization, decoding, envelope-construction, or validation failure returns Unknown. For a seed  $q$ , the logical failure-closed descent follows the deterministic root-to- $q$  chain through

$$D_{\text{eff}} = \max\{0, \min(D_{\text{max}}, D_{\text{tree}}(q) - 1)\}$$

where  $D_{\text{max}}$  is the requested depth cap and  $D_{\text{tree}}(q)$  is the number of available or constructible levels. A linear reference descent returns the first policy-accepted cell; at the cap or a terminal cell, it retains the current domain and status. The evaluated helper instead probes a fixed sparse subset of depths, validating every probed cell independently; it need not find the shallowest or every existing accepted depth. No failed evaluation creates an accepted label. Query-specific endpoint attachments and online corridors remain outside the reusable build and follow Table II.

### Algorithm 1 Zero-gap exact coalescing and budgeted refinement.

---

**Input:** Anchors  $\mathcal{S}$ , LECT  $\mathcal{T}_L$ , obstacles  $\mathcal{O}$ , policy  $V_{\mathcal{O}}$ , descriptor  $\delta$ , depths  $0 \leq D_0 \leq D_s$ , validation cap  $B_{\text{max}} \in \mathbb{Z}_{\geq 1}$ , and resource caps  
**Output:** Labeled boxes  $\mathcal{F}$ , overlap subgraph  $G_{\cap}$ , retained domains  $\mathcal{C}$

- 1: batch-evaluate the identity-deduplicated frontier through depths  $D_0:D_s$ :  
 $s_I \leftarrow \hat{V}_{\mathcal{O}}(I; \delta)$
- 2: exact-coalesce same-revision CertFree groups satisfying ExactUnion
- 3: insert merged  $(B, \text{CertFree})$  and unmerged  $(I, \text{CertFree})$  records into  $\mathcal{F}$ ;  
 build  $G_{\cap}$
- 4: initialize  $\mathcal{P}$  with maximal remaining frontier cells, keyed by region identity and prioritized by anchor coverage;  $\mathcal{C} \leftarrow \emptyset$ ;  $b \leftarrow 0$
- 5: **while**  $\mathcal{P}$  is nonempty,  $b < B_{\text{max}}$ , and time/depth caps remain **do**
- 6:   admit a batch of at most  $B_{\text{max}} - b$  cells;  $b \leftarrow b + |\text{batch}|$
- 7:   evaluate  $s_I \leftarrow \hat{V}_{\mathcal{O}}(I; \delta)$  for each admitted cell
- 8:   **for all** cells  $I$  in the batch **do**
- 9:     **if**  $s_I = \text{CertFree}$  **then**
- 10:       insert  $(I, \text{CertFree})$  into  $\mathcal{F}$  and update  $G_{\cap}$
- 11:     **else if**  $I$  is splittable within the caps **then**
- 12:       enqueue unseen descriptor-aligned children of  $I$
- 13:     **else**
- 14:       defer  $I$  into  $\mathcal{C}$
- 15:     **end if**
- 16:   **end for**
- 17: **end while**
- 18: retain the capacity-limited union of  $\mathcal{C}$  and  $\mathcal{P}$ ; record omissions
- 19: **return**  $\mathcal{F}, G_{\cap}, \mathcal{C}$

---

a) *Finite work:* The reference chain visits at most  $D_{\text{eff}} + 1$  cells and requires  $O(n(D_{\text{eff}} + 1))$  bookkeeping with dimension scans. The evaluated sparse-depth helper probes  $O(1 + \log(D_{\text{eff}} + 1))$  depths. Let  $r$  be the number of root sectors,  $0 \leq D_0 \leq D_s$  the shallow-sweep depths, and  $B_{\text{max}}$  the adaptive-validation cap. An unpruned sweep from  $D_0$  through  $D_s$  validates at most  $r(2^{D_s+1} - 2^{D_0})$  cells, and adaptive refinement adds at most  $B_{\text{max}}$  unique-cell validations. Exact pairwise overlap construction is  $O(nN^2)$  for  $N$  boxes in  $n$  dimensions; replay, envelope construction, and separation remain oracle costs. Finite budgets and bounded OBB-cover loops guarantee termination, but not coverage or completeness.

Here  $G_{\cap} = (\mathcal{F}, E_{\cap})$  is the reusable overlap subgraph; the corridor stage adds the remaining typed edge sets to form  $G$ .

### C. Corridor Query and Certificate

The typed graph separates search adjacency from certifying corridors:  $G = (\mathcal{F}, E_{\cap} \cup E_{\varepsilon} \cup E_s \cup E_{\text{obb}} \cup E_{\text{rep}})$ , comprising overlap/contact, tolerance gap, segment, finite-cover OBB-corridor, and repair edges. Overlap/contact and conservatively validated OBB corridors can support certificate-backed status; other edges do so only after conversion to a conservative corridor. Eligibility is determined by the retained witness, not the edge label. Fix the active scene revision  $\nu_{\text{scene}}$  and descriptor  $\delta$ . Let route provenance be  $W = \langle (t_{j-1}, t_j, R_j) \rangle_{j=1}^m$ , where  $0 = t_0 < \dots < t_m = 1$  and  $R_j$  is the witness region for  $\gamma([t_{j-1}, t_j])$ . Define certificate-backed status by

$$\text{Cert}_{\nu_{\text{scene}}, \delta}(\gamma, W) \iff \text{Contain}(\gamma, W) \wedge \bigwedge_{j=1}^m \text{SceneValid}_{\text{cons}}(R_j; \nu_{\text{scene}}, \delta).$$

Here Contain is the finite subpath-containment condition in Definition 1. The predicate  $\text{SceneValid}_{\text{cons}}(R; \nu_{\text{scene}}, \delta)$  denotes direct conservative validation under that scene revision



and descriptor, or validity inherited by a zero-gap ExactUnion witness. Every box, OBB region, query-endpoint attachment, and transition contributing to the complete path must satisfy Definition 1; one unsupported component downgrades the full return.

Endpoints attach only through containing cells. A best-first traversal uses a heuristic priority and makes no shortest-path claim. If  $N_{\text{adj}}$  adjacency entries are examined,  $N_{\text{rel}}$  relaxations succeed, the heap contains at most  $H$  entries, and each adjacency incurs at most  $T_e$  query-dependent work, the search cost is

$$O(N_{\text{adj}}(1 + T_e) + (N_{\text{rel}} + 1) \log(H + 1)).$$

For overlap routes, gateways lie in consecutive box intersections, and convexity keeps their connecting segments within witnessed boxes.

**Lemma 1** (Box-corridor certificate). *Fix a scene revision  $\nu_{\text{scene}}$  and descriptor  $\delta$ . Let  $\mathcal{F}_{\text{cert}} := \{B_i \mid \sigma_i = \text{CertFree}\}$  contain forest boxes labeled under  $\delta$  against  $\mathcal{O}(\nu_{\text{scene}})$ , each either validated directly by conservative link envelopes or formed through a zero-gap exact-union witness. Define  $R_{\mathcal{F}_{\text{cert}}} := \bigcup_{B \in \mathcal{F}_{\text{cert}}} B$ . If a continuous path  $\gamma : [0, 1] \rightarrow \mathcal{Q}$  satisfies  $\gamma([0, 1]) \subseteq R_{\mathcal{F}_{\text{cert}}}$ , then*

$$\gamma([0, 1]) \subseteq \mathcal{C}_{\text{free}}(\mathcal{O}(\nu_{\text{scene}}))$$

under Assumption 1.

*Proof.* For any  $s \in [0, 1]$ , containment gives  $B \in \mathcal{F}_{\text{cert}}$  with  $\gamma(s) \in B$ . If  $B$  passed the conservative test directly, Corollary 1 applies. If it is an exact coalesced union,  $\gamma(s)$  lies in at least one conservatively validated input box under the same revision and descriptor, so the same corollary applies there. Because  $s$  was arbitrary, the stated path inclusion follows.  $\square$

## VI. EXPERIMENTS

The evaluation addresses four questions. Q1 characterizes warm replay and profile controls on Shelf+KUKA LBR iiwa (Shelf+IIWA); Q2 compares that profile with roadmap, sampling, and convex-region baselines; Q3 studies Medium/Hard random scenes on three robots; and Q4 isolates endpoint- and link-envelope trade-offs. Fig. 3 introduces the fixed Shelf scene. The Q2 baseline couples iterative regional inflation with Graphs of Convex Sets (IRIS/GCS). Table III summarizes the claim and accounting boundary for each question.

a) *Protocol and metrics:* All experiments are software simulations. Runs with retained metadata used Ubuntu 22.04 on an Intel Core i5-12600KF host with 31 GiB of memory and an eight-thread process/library cap; OMPL planners remained single-threaded. Because some imported records omit revision, execution-serialization, or host-load metadata, wall-time comparisons are descriptive. An accompanying machine-readable asset manifest indexes source manifests, hashes, profiles, and selection rules.

Every condition fixes one profile. Build denotes reusable construction after process and cache/database initialization. For RBF, it includes evidence replay/materialization, scene

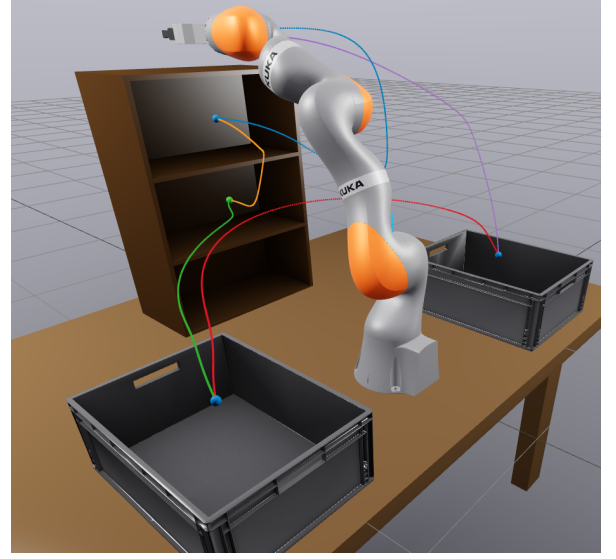


Fig. 3. Canonical Marcucci shelf scene with a KUKA LBR iiwa manipulator (Shelf+IIWA) and fixed anchors: approach-side (AS), target-shelf (TS), center-shelf (CS), left-bin (LB), and right-bin (RB).

labeling, box-graph construction, offline anchors, and accepted offline shortcuts, but excludes snapshot construction and opening. For PRM in Q2 and Q3, saved query endpoints are preloaded outside Build. Online/q is separately timed per-query planner work. The common 0.010-s final simplification and fixed-step path audit ( $\leq 0.01$  rad in the native-joint metric, with zero collision tolerance) are excluded from Build, Online/q, and scene-batched task time per query (Task/q); path ratios use the postprocessed, audit-passing return. The displayed trade-offs are therefore within-method accounting curves.

The Q1 Batch-task values and Q3 Task/q values are scene-batched diagnostics. For a Q3 scene with  $N_q$  saved queries,

$$\text{Task/q} = \left( T_{\text{adapt}}^{\text{wall}} + \sum_i T_{\text{query},i}^{\text{internal}} \right) / N_q,$$

where the terms denote scene-level adaptation and query-internal planning time. Build/ $K$  plus the displayed query/task coordinate uses  $K$  as a reuse horizon; values beyond the five Shelf or ten random-scene queries are algebraic projections, not additional runs.

Unless stated otherwise, entries are success-only medians [first quartile, third quartile]. For waypoints  $q_0, \dots, q_m$ ,  $L = \sum_{j=0}^{m-1} \|q_{j+1} - q_j\|_2$ . Shelf ratios  $L_q/L_{\text{ref},q}$  and random-scene ratios  $L/L_{\text{ref},\lambda}$  use the shortest audited return for the exact saved query in the corresponding reported method/checkpoint pool. Reference lengths are fixed over that full pool before operating-point selection. These query-matched references are empirical normalizers. All planning rows use the collision model in Assumption 1.

b) *Profile inheritance and cache accounting:* Q1 uses the read-only full-root d23 LECT configuration, and Q2 inherits Q1's b100 Shelf profile. Q3 retains the partition-native backend and common audit and final-simplification policies, but deliberately does not inherit b100: it uses six robot-

TABLE III  
EVALUATION QUESTIONS, EVIDENCE, AND ACCOUNTING BOUNDARIES UNDER TABLE II

| Question            | Tested claim                                         | Main evidence                                                                                    | Boundary emphasized                                                        |
|---------------------|------------------------------------------------------|--------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------|
| Q1 Shelf profile    | Characterize warm replay and profile controls.       | b100 (100-box budget), d23 (LECT depth 23), replay/no-replay controls, and the five-query cycle. | Batch task is diagnostic; residual transitions remain validation-required. |
| Q2 Shelf comparison | Compare reusable-build and path-quality points.      | Eight-seed IRIS/GCS, PRM, RRT-Connect, and BIT* context.                                         | Method-specific reusable setup precludes a shared latency ranking.         |
| Q3 Random scenes    | Evaluate audited multi-robot paths and path quality. | Six 100-query Medium/Hard catalog-tuned conditions.                                              | The 100/100 audit requirement is in-catalog selection eligibility.         |
| Q4 Mechanisms       | Isolate endpoint- and link-envelope trade-offs.      | Shared deterministic 1000-box sweeps at six widths.                                              | Mechanism costs are not end-to-end planning latency.                       |
| Archived diagnostic | Record scene-cache maintenance primitives.           | Historical insertion, removal, and warm rebuild.                                                 | Outside Q1–Q4.                                                             |

difficulty profiles selected on the frozen catalog and held fixed across ten seeds. Their complete parameter vectors are in the source manifests indexed by the asset manifest. Q3 uses robot-local, query-independent anchors. Its IIWA profile reuses only the full-domain, kinematics-only d23 snapshot—no Shelf labels, edges, or anchors—whereas UR5 and Panda use their registered prewarmed stores. UR5 Medium keeps added edges query-local; the other five profiles reuse them within a scene.

Warm-cache Build excludes snapshot creation/opening, storage, and peak memory, but includes lookup, evidence reads, and demand paging. The d23 snapshot has 16,777,215 records, occupies 12.74 GiB, and took 1008.6 s to prewarm. In Q1, the baseline, Link-AABB, and SupportHull-only rows use the full-root d23 IFK-AA snapshot; the last two hold endpoint replay fixed while varying the link test. Critical-sample uses descriptor-matched d23 evidence, whereas HIFK-5 disables external replay and materializes endpoint evidence on demand. Q1 therefore uses compound profile controls; Q4 isolates endpoint- and link-envelope costs.

#### A. Q1: Shelf+IIWA Replay and Profile Controls

Q1 characterizes the d23 warm baseline and its replay/no-replay controls. Fig. 3 shows the fixed five-anchor cycle, executed as AS–TS–CS–LB–RB–AS. Table IV reports the b100 profile over seeds 0–7; b100 was development-selected on this cycle, and its complete parameter vector is in the indexed Q1 source manifest.

The cached IFK-AA/SupportHull baseline builds in a median 0.171 s; on-demand HIFK-5 materialization takes 2.770 s. Its 0.077 s Batch-task value is completion time within the five-query bridge batch and includes shared bridge work, so it is a batch diagnostic.

All 40 baseline paths pass the 0.01-rad audit and retain validation-required status under the reporting contract. Link-AABB changes median task time from 0.077 s to 0.083 s and median  $L_q/L_{ref,q}$  from 1.16 to 1.21; removing the AABB broadphase changes the five-query amortized diagnostic from 0.111 s to 0.126 s.

*Takeaway.* Warm d23 replay builds 16.2 times faster than on-demand HIFK-5 in this compound profile; Q4 isolates endpoint-construction cost.

#### B. Q2: Shelf Cross-Algorithm Context

Q2 compares Q1’s b100 profile with four baselines on the same five queries. RRT-Connect and BIT\* run independently per query; IRIS/GCS and PRM use reusable, query-set-informed structures. PRM shares one endpoint-preloaded cumulative roadmap per seed and selects its 20-s checkpoint. The indexed source manifests retain all caps, checkpoints, and structural parameters. Operating points are selected on the five queries across eight stochastic runs per method.

For each query, PRM and BIT\* select the full-seed-success checkpoint with minimum median planner time  $T$  among those within 1% of that query’s best recorded mean audited path length. IRIS/GCS+post selects the fastest prefix with 40/40 final-audit success and aggregate mean path ratio within 8% of its recorded best, yielding r8. The figure uses method-level curves and each baseline’s first display point within 8% of its best path. At r8, native/final audit counts are 0/40 and 40/40 because accepted IRIS/GCS returns use the common postprocessor; the final paths determine path statistics while postprocessing is excluded from  $T$ .

*Takeaway.* At the selected Shelf points, RBF has lower query-wise path-ratio medians than PRM and RRT-Connect for all five queries, lower than IRIS/GCS+post for one, and higher than BIT\* for all five.

#### C. Q3: Random Multi-Robot Scenes

Q3 characterizes audited paths and path quality for six catalog-tuned Medium/Hard profiles across three robots.

*Dataset and query construction.* The frozen catalog contains three robots, ten seeds per difficulty, and ten queries per scene. Q3 evaluates all 60 Medium/Hard scenes (600 evaluations): 300 start–goal pairs shared across paired obstacle prefixes, with Hard strictly extending Medium. Every direct start–goal segment is obstructed. A fixed planner-probe policy conditioned catalog admission on obstruction and feasibility; every method was then rerun on the complete admitted set.

*Compared methods and accounting.* RBF reuses one query-agnostic forest across each scene’s ten saved-order queries. The six profiles were selected on this catalog for 100/100 audit eligibility using Task/q and path length. The Q3 PRM/BIT\* baselines retain Q2’s structural profiles with manifested horizon overrides. PRM shares one endpoint-preloaded roadmap per scene. RRT-Connect and BIT\* run independently per

TABLE IV  
SHELF+IIWA RBF BASELINE AND PROFILE-LEVEL CONTROLS UNDER THE d23 SCHEDULE

| Case             | Build                | Batch task           | Diag. 5q             | $L_q / L_{ref,q}$ |
|------------------|----------------------|----------------------|----------------------|-------------------|
| RBF baseline     | 0.171 [0.166, 0.174] | 0.077 [0.072, 0.085] | 0.111 [0.107, 0.117] | 1.16 [1.09, 1.40] |
| Critical-sample  | 0.279 [0.275, 0.283] | 0.087 [0.078, 0.095] | 0.145 [0.134, 0.151] | 1.16 [1.10, 1.38] |
| Link-AABB        | 0.167 [0.164, 0.171] | 0.083 [0.077, 0.086] | 0.116 [0.111, 0.119] | 1.21 [1.09, 1.50] |
| HIFK-5 endpoint  | 2.770 [2.644, 3.003] | 0.181 [0.139, 0.207] | 0.739 [0.703, 0.777] | 1.16 [1.10, 1.44] |
| SupportHull-only | 0.193 [0.188, 0.200] | 0.087 [0.082, 0.095] | 0.126 [0.120, 0.136] | 1.16 [1.09, 1.40] |

Notes: Seconds; medians [first quartile, third quartile]. Build summarizes eight seed observations; other columns summarize 40 seed–query observations. Replay is full-root d23 IFK-AA, matched d23 Critical-sample, or disabled for HIFK-5. Batch task is in-batch; Diag. 5q = Build/5 + Batch task.

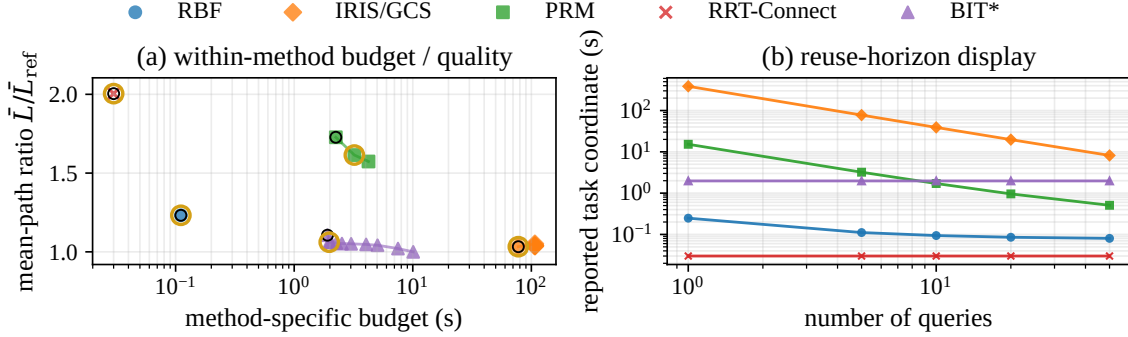


Fig. 4. Within-method Shelf+IIWA accounting curves over eight seeds. Coordinates are method-specific: RBF Build/5+batch task; IRIS/GCS and PRM Build/5+query; RRT-Connect query; and BIT\* measured query work per checkpoint. Abscissae exclude final postprocessing/audit. Ordinates normalize each aggregate success-only mean audited path length by the shortest such displayed mean. Black rings mark the first point at which all five queries pass audit for all eight seeds. Gold rings mark the reported RBF point or the first baseline point within 8% of that method’s best recorded mean path. PRM/BIT\* display markers need not match the query-wise selectors in Table V. Panel (b) varies the scene-build reuse horizon.

TABLE V  
SHELF+IIWA CROSS-ALGORITHM COMPARISON OF AUDITED PATHS

| Query | RBF b100<br>Build 0.171 s |                      | IRIS/GCS+post<br>Build 385.950 s |                      | PRM<br>Build 20.040 s   |                      | RRT-Connect<br>No reusable build |                      | BIT*<br>No reusable build  |                      |
|-------|---------------------------|----------------------|----------------------------------|----------------------|-------------------------|----------------------|----------------------------------|----------------------|----------------------------|----------------------|
|       | Batch task (s)            | $L_q / L_{ref,q}$    | $T$ (s)                          | $L_q / L_{ref,q}$    | $T$ (s)                 | $L_q / L_{ref,q}$    | $T$ (s)                          | $L_q / L_{ref,q}$    | $T$ (s)                    | $L_q / L_{ref,q}$    |
| AS→TS | 0.065<br>[0.064, 0.070]   | 1.40<br>[1.34, 1.65] | 0.540<br>[0.390, 0.661]          | 1.15<br>[1.11, 1.18] | 0.198<br>[0.147, 0.248] | 1.61<br>[1.42, 1.73] | 0.034<br>[0.024, 0.052]          | 2.27<br>[2.05, 2.72] | 4.004<br>[4.002, 4.004]    | 1.06<br>[1.02, 1.08] |
| TS→CS | 0.085<br>[0.082, 0.089]   | 1.13<br>[1.11, 1.14] | 0.346<br>[0.296, 0.396]          | 1.22<br>[1.11, 1.28] | 0.173<br>[0.118, 0.312] | 2.99<br>[2.11, 3.22] | 0.054<br>[0.021, 0.089]          | 3.98<br>[2.80, 5.09] | 10.138<br>[10.004, 10.386] | 1.09<br>[1.04, 1.11] |
| CS→LB | 0.073<br>[0.070, 0.077]   | 1.50<br>[1.35, 1.85] | 0.411<br>[0.314, 0.518]          | 1.03<br>[1.02, 1.05] | 0.136<br>[0.120, 0.195] | 1.53<br>[1.42, 2.13] | 0.032<br>[0.027, 0.070]          | 2.24<br>[1.79, 2.56] | 7.504<br>[7.504, 7.725]    | 1.05<br>[1.04, 1.08] |
| LB→RB | 0.078<br>[0.075, 0.083]   | 1.10<br>[1.08, 1.24] | 0.761<br>[0.685, 0.879]          | 1.02<br>[1.01, 1.06] | 0.164<br>[0.091, 0.399] | 1.20<br>[1.11, 1.29] | 0.001<br>[0.001, 0.002]          | 1.31<br>[1.29, 1.34] | 1.654<br>[1.654, 1.654]    | 1.03<br>[1.03, 1.04] |
| RB→AS | 0.081<br>[0.077, 0.086]   | 1.05<br>[1.04, 1.05] | 0.537<br>[0.418, 0.609]          | 1.00<br>[1.00, 1.00] | 0.252<br>[0.096, 0.295] | 1.24<br>[1.18, 1.25] | 0.001<br>[0.001, 0.001]          | 1.08<br>[1.04, 1.10] | 0.204<br>[0.204, 0.204]    | 1.02<br>[1.01, 1.03] |

Notes: Seconds; medians [first quartile, third quartile] over eight seeds. RBF Batch task is in-batch; other  $T$  values are timed per query. Coordinates exclude final postprocessing/audit; selectors follow Q2.

query; BIT\* selects the smallest audit-passing solve coordinate within 8% of each saved query’s best path. For PRM, curve abscissae use configured checkpoints, whereas the table orders 100/100 points by measured Build/5+query and advances only after each  $\geq 1\%$  aggregate improvement. BIT\* curve rings also track successive  $\geq 1\%$  aggregate improvements, whereas table entries use the query-wise 8% rule above. IRIS/GCS is omitted because no common-protocol Q3 result was available.

**Results.** At the six eligible points, RBF medians are 0.02–0.31 below RRT-Connect and lower than PRM in five conditions. They are 0.01–0.05 below BIT\* in five conditions and 0.01 higher for Panda-Medium. Mean RBF Build/ $K$ +Task/ $q$  falls 70–79% from  $K = 1$  to  $K = 10$ .

#### D. Q4: Endpoint and Link-Envelope Mechanisms

Q4 reports six-width joint-interval sweeps in Tables VII and VIII. Each study uses the same deterministic set of 1000 IIWA joint boxes per width for all representations. The endpoint study compares construction time and volume. Its Gap metric is the negative of the largest per-axis extent-width shortfall relative to the union of Critical-sample, Analytical, and Monte Carlo extents; it is a span diagnostic. Endpoint entries are medians and worst-case Gap, whereas link entries are means.

The link study isolates representation cost, not the production AABB-first cascade. For each box and representation, Test pools five timing repeats in two cases: the Shelf obstacle set

TABLE VI  
MEDIUM/HARD RANDOM-SCENE SUMMARIES AT THE REPORTED OPERATING POINTS

| Scenario     | RBF            |                |                     | PRM            |                |                     | RRT-Connect    |                     | BIT*           |                     |
|--------------|----------------|----------------|---------------------|----------------|----------------|---------------------|----------------|---------------------|----------------|---------------------|
|              | Build          | Task/q         | $L/L_{ref,\lambda}$ | Build          | Online/q       | $L/L_{ref,\lambda}$ | Online/q       | $L/L_{ref,\lambda}$ | Selected $T/q$ | $L/L_{ref,\lambda}$ |
| IIWA-Medium  | 0.075          | 0.009          | 1.00                | 3.567          | 0.038          | 1.03                | 0.002          | 1.02                | 0.014          | 1.05                |
|              | [0.058, 0.083] | [0.007, 0.011] | [1.00, 1.03]        | [3.007, 3.933] | [0.014, 0.070] | [1.02, 1.12]        | [0.001, 0.002] | [1.01, 1.05]        | [0.007, 0.042] | [1.03, 1.07]        |
| IIWA-Hard    | 0.097          | 0.014          | 1.00                | 4.175          | 0.048          | 1.03                | 0.002          | 1.03                | 0.027          | 1.05                |
|              | [0.086, 0.105] | [0.011, 0.015] | [1.00, 1.03]        | [3.408, 5.637] | [0.021, 0.102] | [1.02, 1.08]        | [0.002, 0.003] | [1.00, 1.07]        | [0.012, 0.076] | [1.03, 1.07]        |
| UR5-Medium   | 0.262          | 0.075          | 1.05                | 3.124          | 0.323          | 1.02                | 0.023          | 1.23                | 0.026          | 1.06                |
|              | [0.254, 0.270] | [0.020, 0.086] | [1.00, 1.14]        | [2.966, 3.418] | [0.085, 0.426] | [1.00, 1.12]        | [0.002, 0.049] | [1.06, 1.54]        | [0.012, 0.058] | [1.03, 1.14]        |
| UR5-Hard     | 0.276          | 0.083          | 1.05                | 2.488          | 0.419          | 1.06                | 0.048          | 1.36                | 0.046          | 1.07                |
|              | [0.264, 0.284] | [0.043, 0.108] | [1.01, 1.14]        | [2.325, 3.048] | [0.129, 0.437] | [1.01, 1.19]        | [0.008, 0.090] | [1.14, 1.58]        | [0.022, 0.091] | [1.03, 1.17]        |
| Panda-Medium | 0.250          | 0.028          | 1.05                | 3.431          | 0.066          | 1.06                | 0.005          | 1.22                | 0.016          | 1.04                |
|              | [0.218, 0.265] | [0.013, 0.039] | [1.02, 1.09]        | [3.364, 4.180] | [0.038, 0.159] | [1.03, 1.08]        | [0.002, 0.011] | [1.12, 1.35]        | [0.007, 0.031] | [1.02, 1.06]        |
| Panda-Hard   | 0.224          | 0.043          | 1.05                | 2.337          | 0.054          | 1.10                | 0.056          | 1.25                | 0.062          | 1.06                |
|              | [0.221, 0.237] | [0.021, 0.053] | [1.01, 1.16]        | [1.627, 3.978] | [0.034, 0.126] | [1.05, 1.16]        | [0.019, 0.136] | [1.14, 1.37]        | [0.027, 0.124] | [1.04, 1.12]        |

Notes: Seconds; medians [first quartile, third quartile]. Build and RBF Task/q summarize 10 scenes; query and ratio fields summarize 100 queries. Selectors follow Q3; UR5 Medium uses query-local edges.  $\lambda$  identifies the exact scene-query.

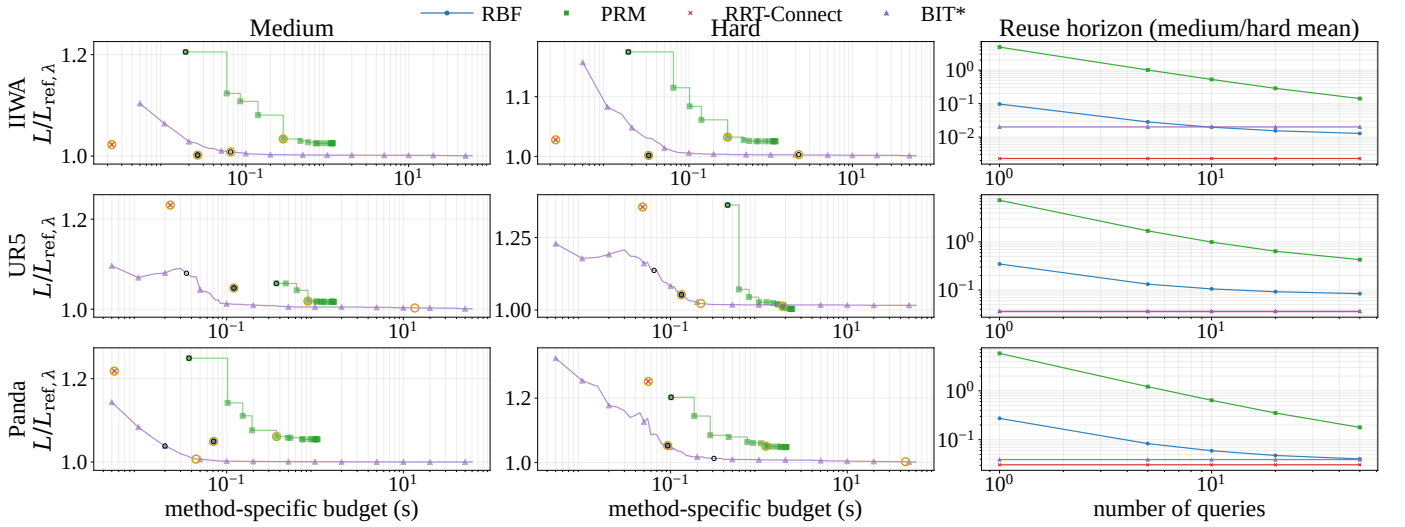


Fig. 5. Random-scene within-method accounting and path quality for six conditions. Left ( $K = 5$ ): RBF Build/ $K$ +Task/q, PRM checkpoint/ $K$ +query, RRT-Connect query, and BIT\* checkpoint. Coordinates exclude final postprocessing/audit and are not cross-method latencies. Black rings mark first 100/100 trace points; gold rings mark the reported RBF/RRT-Connect points or the last PRM/BIT\* point retained after successive  $\geq 1\%$  ratio improvements. Right: for each robot, the mean of the selected Medium and Hard coordinates versus  $K$ . Ratios use exact query-matched references; table selectors differ as stated in Q3. Pre-black-ring points are success-only survivor summaries and are ineligible for table selection.

translated 20 m along the  $x$ -axis, and a method-specific 10-mm known-overlap AABB centered in that representation's first link envelope. Both cases scan all link-obstacle pairs without early exit. Link AABB uses overlap tests; SupportHull invokes GJK with its AABB broadphase disabled. Thus the measurements quantify envelope-build and narrow-phase test costs and motivate the production choice of cheap AABB rejection before optional SupportHull/GJK refinement.

*Takeaway.* At 0.50 rad, HIFK-5 has normalized endpoint volume 2.07 versus 8.61 for IFK-AA (76% lower) at 9.8 times the endpoint-construction cost. Across the tested widths, isolated AABB overlap takes 4.7–6.1% of the mean per-probe time required by direct SupportHull/GJK. Q4 isolates the mechanisms whose system-level interaction appears in Q1.

#### E. Archived Scene-Cache Maintenance Diagnostic

Table IX reports maintenance-only scene-cache timing.

TABLE VII  
ENDPOINT-ENVELOPE EVALUATION OVER JOINT-INTERVAL SIDE LENGTHS (RADIANs)

| Source          | Volume ratio |      |      |      |      |      | Time ( $\mu$ s) |         |          |          |          |          | Gap (mm) |        |        |        |        |         |
|-----------------|--------------|------|------|------|------|------|-----------------|---------|----------|----------|----------|----------|----------|--------|--------|--------|--------|---------|
|                 | 0.02         | 0.05 | 0.10 | 0.20 | 0.30 | 0.50 | 0.02            | 0.05    | 0.10     | 0.20     | 0.30     | 0.50     | 0.02     | 0.05   | 0.10   | 0.20   | 0.30   | 0.50    |
| IFK-AA          | 1.09         | 1.23 | 1.52 | 2.30 | 3.53 | 8.61 | 5.48            | 6.11    | 6.45     | 6.58     | 6.81     | 7.35     | –        | –      | –      | –      | –      | –       |
| HIFK-3          | 1.04         | 1.10 | 1.22 | 1.51 | 1.89 | 3.12 | 25.94           | 26.85   | 29.16    | 27.21    | 29.27    | 29.38    | –        | –      | –      | –      | –      | –       |
| HIFK-5          | 1.02         | 1.07 | 1.13 | 1.30 | 1.51 | 2.07 | 70.40           | 71.69   | 73.82    | 73.15    | 75.92    | 72.32    | –        | –      | –      | –      | –      | –       |
| Critical-sample | 1.00         | 1.00 | 1.00 | 1.00 | 0.99 | 0.98 | 8.68            | 9.04    | 10.03    | 11.56    | 12.69    | 16.33    | -0.05    | -0.30  | -1.17  | -4.53  | -10.77 | -30.97  |
| Analytical      | 1.00         | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 6429.96         | 6500.41 | 6602.25  | 6800.10  | 6927.12  | 7096.01  | 0.00     | 0.00   | 0.00   | 0.00   | 0.00   | -4.49   |
| Monte Carlo     | 0.64         | 0.70 | 0.74 | 0.78 | 0.79 | 0.82 | 2251.74         | 5629.24 | 11233.26 | 22352.95 | 33813.58 | 56835.57 | -6.05    | -14.62 | -32.68 | -65.57 | -88.87 | -123.53 |

Notes: 1000 shared boxes/width; medians; volume/Analytical. Monte Carlo uses  $\text{round}[(50000/0.35)w]$  samples/box (2857–71429). Gap is worst extent-width shortfall; – is unreported. IFK-AA/HIFK are certificate-capable under the stated conditions.

TABLE VIII  
LINK-ENVELOPE EVALUATION OVER JOINT-INTERVAL SIDE LENGTHS (RADIANs)

| Envelope    | Core volume ( $\text{m}^3$ ) |         |        |       |       |      | Build ( $\mu$ s) |       |       |       |       |       | Test ( $\mu$ s) |        |        |        |        |        |
|-------------|------------------------------|---------|--------|-------|-------|------|------------------|-------|-------|-------|-------|-------|-----------------|--------|--------|--------|--------|--------|
|             | 0.02                         | 0.05    | 0.10   | 0.20  | 0.30  | 0.50 | 0.02             | 0.05  | 0.10  | 0.20  | 0.30  | 0.50  | 0.02            | 0.05   | 0.10   | 0.20   | 0.30   | 0.50   |
| Link AABB   | 0.0136                       | 0.0201  | 0.0384 | 0.147 | 0.508 | 4.61 | 0.138            | 0.152 | 0.167 | 0.166 | 0.171 | 0.178 | 0.586           | 0.605  | 0.601  | 0.604  | 0.630  | 0.790  |
| SupportHull | 0.000359                     | 0.00266 | 0.0143 | 0.106 | 0.449 | 4.54 | 0.205            | 0.214 | 0.218 | 0.219 | 0.217 | 0.250 | 12.520          | 12.850 | 12.267 | 12.223 | 12.659 | 13.026 |

Notes: 1000 shared boxes/width; entries are means. Test pools five free and known-overlap probes without early exit. Link AABB uses overlap; SupportHull uses direct GJK. Build excludes endpoint envelopes.

TABLE IX  
TIMING OF ARCHIVED SCENE-CACHE MAINTENANCE PRIMITIVES

| Operation                  | Time (ms)                | Warm/update           |
|----------------------------|--------------------------|-----------------------|
| Warm build (10 obstacles)  | 922.75 [897.57, 955.67]  | –                     |
| Insert 10 $\rightarrow$ 15 | 6.06 [5.69, 6.54]        | 153.3–207.5 $\times$  |
| Remove 15 $\rightarrow$ 10 | 0.35 [0.31, 1.65]        | 943.6–3354.0 $\times$ |
| Warm build (15 obstacles)  | 958.79 [941.67, 1115.18] | –                     |

Notes: Historical; excluded from Q1–Q4. Values are median [first quartile, third quartile] milliseconds; Warm/update is the paired target-state warm-build/update ratio (15 obstacles for insertion; 10 for removal).

## VII. CONCLUSION

RBF combines two-level volumetric reuse with a failure-closed route-status contract. Within a scene, it reuses a budgeted graph of joint boxes with zero-gap exact-union coalescing. Across compatible revisions, direct-store LECT replay admits only identity-matched endpoint evidence; the evaluated snapshot assumes an immutable, co-generated image of its verified companion store. Link envelopes are re-derived, and scene labels and connectivity must be rebuilt under the current descriptor. Retained typed witnesses separate search from certificate-backed connectivity.

On Shelf+IIWA, warm IFK-AA replay and on-demand HIFK-5 materialization yield median builds of 0.171 s and 2.770 s, respectively—a 16.2-fold compound-profile difference. Across six in-catalog Q3 conditions with method-specific operating-point selection, RBF’s query-matched median path ratios span 1.00–1.05, versus 1.02–1.36 for RRT-Connect and 1.04–1.07 for BIT\*. Amortizing Build over ten queries lowers RBF’s Medium/Hard mean build-plus-task diagnostic by 70–79% relative to one query. At 0.50 rad, HIFK-5 reduces normalized endpoint volume by 76% at 9.8 times the IFK-AA construction cost; isolated AABB tests take 4.7–6.1% of direct SupportHull/GJK probe time.

*Limitations.* Evaluation is software-only and uses conditioned points without held-out Q3 selection. The Q1/Q3 ar-

tifacts lack complete final-route provenance, and the experiments do not test online revision or cache mismatch. The collision model covers active rounded links against AABB obstacles. Finite depth and bridge/repair budgets forfeit completeness, the 12.74-GiB complete d23 tree scales exponentially, and certificate status remains conditional on the stated floating-point contract.

RBF therefore provides an auditable volumetric reuse layer that complements roadmap reuse and convex-region optimization.

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